

Flatten a Nested List

Write a Python function to flatten a nested list.

Input: [[1, 2], [3, 4], [5]]

Output: [1, 2, 3, 4, 5]

```
def flatten_list(nested_list):
    result = []

    for sublist in nested_list:
        for item in sublist:
            result.append(item)

    return result
```

```
numbers = [[1, 2], [3, 4], [5]]
```

```
print(flatten_list(numbers))
```

Output : [1, 2, 3, 4, 5]

Merge Two Sorted Lists

Write a Python function to merge two sorted lists into a single sorted list.

Input: [1, 3, 5], [2, 4, 6]

Output: [1, 2, 3, 4, 5, 6]

```
def merge_sorted_lists(list1, list2):
    result = []
    i = 0
    j = 0

    while i < len(list1) and j < len(list2):
        if list1[i] < list2[j]:
            result.append(list1[i])
            i += 1
        else:
            result.append(list2[j])
            j += 1

    while i < len(list1):
        result.append(list1[i])
        i += 1

    while j < len(list2):
```

```
result.append(list2[j])
```

```
j += 1
```

```
return result
```

```
list1 = [1, 3, 5]
```

```
list2 = [2, 4, 6]
```

```
print(merge_sorted_lists(list1, list2))
```

```
Output : [1, 2, 3, 4, 5, 6]
```

Find All Pairs in a List that Sum to a Specific Value

Write a Python function to find all pairs in a list that sum to a specific value.

Input: [1, 2, 3, 4, 5], Sum=6

Output: [(1, 5), (2, 4)]

```
def find_pairs(numbers, target):
```

```
    pairs = []
```

```
    for i in range(len(numbers)):
```

```
        for j in range(i + 1, len(numbers)):
```

```
            if numbers[i] + numbers[j] == target:
```

```
                pairs.append((numbers[i], numbers[j]))
```

```
    return pairs
```

```
numbers = [1, 2, 3, 4, 5]
```

```
target = 6
```

```
print(find_pairs(numbers, target))
```

```
Output: [(1, 5), (2, 4)]
```